Fiscal Year:	FY 2013	**Task Last Updated:**	FY 08/08/2013
PI Name:	Chaikin, Paul M. Ph.D.		
Project Title:	The Control and Dynamics of Hard Sphere Colloidal Dispersions		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	COMPLEX FLUIDS/SOFT MATTER--Complex Fluids		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	chaikin@NYU.edu	**Fax:**	FY 212 995-4016
PI Organization Type:	UNIVERSITY	**Phone:**	212 998-7694
Organization Name:	New York University		
PI Address 1:	Department of Physics		
PI Address 2:			
PI Web Page:			
City:	New York	**State:**	NY
Zip Code:	10003	**Congressional District:**	8
Comments:	NOTE: PI moved to NYU (from Princeton U) in 2005 per A. Hollingsworth in PI's dept (7/2009). Changed email 3/30/2009 (chaikin@princeton.edu no longer valid).		
Project Type:	Flight	**Solicitation / Funding Source:**	98-HEDS-03
Start Date:	06/01/2008	**End Date:**	09/05/2013
No. of Post Docs:	1	**No. of PhD Degrees:**	3
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA GRC
Contact Monitor:	Motil, Brian J.	**Contact Phone:**	216-433-6617
Contact Email:	brian.j.motil@nasa.gov		
Flight Program:	Shuttle/ISS		
Flight Assignment:	ISS NOTE: End date changed to 9/5/2013 per NSSC information (Ed., 3/10/14) NOTE: Extended to 5/31/14 per NSSC information (Ed., 8/5/13)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Hollingsworth, Andrew (New York University)		
Grant/Contract No.:	NNX08AK04G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Colloid science is entering a new era. Over the past 15 years, our NASA-sponsored research has mainly dealt with monodisperse suspensions of colloidal particles interacting via well-known forces. Using spherical particles and observations with light scattering and microscopy, we have gained a great deal of fundamental knowledge about different phases of matter and the dynamics and thermodynamics of their formation. In particular, our experimental results in microgravity have lead to a basic understanding of why crystals and glasses form and their properties. During the past decade, we have made great strides in synthesizing new classes of particles with different shapes and specific, reversible or irreversible, variable range interactions. We have also found new ways to manipulate the particles with flow, electric and magnetic fields, and light. We are therefore positioned at the threshold of a new technology, assembling equilibrium and non-equilibrium macroscopic structures with function and activity from well designed particles on the nano to micron scale. Of course, there are still fundamental scientific questions which we can and will address including a host of new ordered phases, frozen configurations, frustration and glasses and the process of self-organization itself. In particular, we plan to use the microscopy and light scattering instruments, in collaboration with our European colleagues, to study particles that we prepare through emulsion and dispersion polymerization. Physical lithographic techniques will also be employed, and the particles will be modified chemically for controllable interactions. We plan to use different phoretic techniques– electro-, dielectro-, and thermo-phoresis– to control the particles density and orientation. These will also serve as the driving forces to establish the rheological properties of these new systems. EDITOR'S NOTE (4/07/2014): PROJECT CONTINUES AS "The Control and Dynamics of Hard Sphere Colloidal Dispersions--NNX13AR67G" WITH THE SAME PRINCIPAL INVESTIGATOR; PERIOD OF PERFORMANCE IS 9/6/2013-9/5/2018. SEE THAT PROJECT FOR SUBSEQUENT REPORTING.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Characterization of crystal formation in the microgravity environment of the ISS (International Space Station) can lead to a greater understanding of how gravity affects many kinds of colloidal materials, including monodisperse ellipsoids and cubes, colloidal clusters of silica or polymer microspheres, DNA-functionalized colloidal spheres, and 'lock-and-key' colloids. By performing these experiments in reduced gravity, we intend to accomplish the desired characterization without gravitationally-induced inhomogeneities that affect both the dynamics and equilibrium state on Earth. Understanding these complex materials should enable new ways of forming ordered phases, such as those sought for photonic devices to be used in optical communication systems. With the ability to make particles of different shapes, i.e., non spherical, we also have the possibility of having directionally dependent particle interactions. For example, we could take tetrahedral clusters of particles and attach DNA to them. The complementary single-stranded DNA 'sticky ends' can associate/dissociate via thermal activation. This arrangement could lead to tetrahedral bonding as found in diamond or in amorphous glass structures. Another approach utilizes depletion interactions. Since we can lithographically prepare particles of any shape we design in two dimensions and many shapes in three dimensions, we can fabricate lock-and-key colloids which only bind to their complementary shape. In this case, the binding is also directional since the congruent surfaces must match. We can also make such lock-and-key particles through emulsion chemistry. Our goal is to produce some simple processes with such ‘designer particles’ and interactions, to lay the foundations for self-assembly and perhaps self-replication of this new class of materials.
	At the beginning of the last funding cycle we could only dream of the type of particles that have been fabricated and are now in hand ready for ground based research and some already scheduled for flight. Among the new particles are combinations of polymer[1], silica[2] and hematite[3] particles of different shapes and functionalities. They are made from different advanced chemical synthesis[4,5], from different lithographic techniques[6], from emulsions[7], microfluidics and different functionalizations[8]. Our most advanced flight ready particles, set for flight in the coming, year are polymer cubes, actually “supercubes”, with different corner roundings and described by the equation: 1 = (x/a)m+ (y/a)m+(z/a)m, where m varies from 2 to 4. The usual cubes, m=2, pack most densely in a cubic lattice. The cubes with rounded edges pack most densely in a tilted lattice taking advantage of the space at their corners[9]. Most interestingly using depletents of different sizes we can fill the edges and corners of the cubes and change the packing/crystal structure[5]. During this period we have also achieved our goal of creating “lock and key” particles[4] which can specifically recognize a complementary particle by shape in analogy with the cartoon pictures of how proteins recognize their complements. The spherical particles have a partially spherical “mouth” like a PACMAN particle into which a particle of the correct radius will fit. These particles can be assembled by adding depletent into dimers, trimmers, tetramers etc. as well as complex chains and polymers. We also have patchy particles of different varieties that can bind to other particles only at specific angles. In one case these particles have “bumps” on their surface in the configurations of dimers, tetramers etc[10]. In another case they can have arbitrary patches with different functionalities formed by different DNA patches templated on their surfaces or written by photolithography[11]. In turn the bumps or patches can be functionalized as DNA sticky ends or as hydrophobic or hydrophilic regions. Over the past several years a number of groups, including our own under NASA support[15], have demonstrated that different phoretic effects, dielectrophoresis, thero-phoresis[16] and difusiophoresis[17] can be used to drive particles in quasi-equilibrium into well ordered crystalline states. Osmosis and phoresis are complementary effects, in a gradient of electrostatic potential, temperature or concentration, there will be an osmotic fluid flow on the surface of a particle held at fixed position in space. If the particle is released it will move phoretically in the opposite direction driven by that flow. Under partial NASA support we have recently shown that such self-propelled osmotically driven particles collide and form dynamic “living crystals”[18]. A unique aspect of our swimming particles is that they were light activated so that the swimming could readily be turned on and off by application of blue light so that we could contrast the effects of thermal diffusion with that of self-driven motion. Again it will be quite interesting to combine thermodynamic interactions, such as depletion, with out of equilibrium driven particle motion. We also want to explore whether such thermodynamic interactions as depletion related to diffusion of the depletant can be substantially enhanced by using active depletants which can exhibit an “effective” temperature many orders of magnitude higher than the thermodynamic temperature. We have also found that the well known depletion effect has some previously undiscovered attributes. Depletion is an interaction that is proportional to temperature but always attractive. When depletant particles absorb on a larger particle

Task Progress:	they can bridge to another large particle again returning an attractive interaction. The literature suggests that with depletant particles always attract (unless the depletant is too small)[12]. We have discovered that there is an in between region where absorption cancels depletion and clusters dissolve. In fact we start at one temperature with crystalline aggregates, heat the system and the crystals melt. We then heat further and cluster aggregates form. This gives a unique control of the properties of the system by varying temperature. References: 1. Pelton, R. H. & Chibante, P. Preparation of aqueous lattices with n-isopropylacrylamide. Colloids Surf. 20, 247–256. 2. Stober, W., Fink, A. & Bohn, E. Controlled growth of monodisperse silica spheres in micron size range. J. Colloid Interf. Sci. 26, 62–69 (1968). 3. T. Sugimoto, M. Khan and A. Maramatsu, Colloids Surf., A, 1993, 70, 167–169. (preparation of hematite cubes. 4. S. Sacanna, W. T. M. Irvine, P. M. Chaikin1 D. J. Pine, “Lock and key colloids”, Nature 464, 575-578 (2010). 5. Laura Rossi, Stefano Sacanna, William T. M. Irvine, Paul M. Chaikin, David J. Pine and Albert P. Philipse, “Cubic crystals from cubic colloids”, Soft Matter, , 7, 4139 (2011). 6. Feng L, Romulus J, Li M, Sha R, Royer J, Wu KT, Xu Q, Seeman NC, Weck M, Chaikin P., “Cinnamate-based DNA photolithography”, Nature Materials. 2013 May 19. http://dx.doi.org/ . 7. Pontani, L.-L., Jorjadze, I., Viasnoff, V. and Brujic, J. Biomimetic emulsions reveal the effect of mechanical forces on cell–cell adhesion Proc. Natl Acad. Sci. USA 109, 9839-9844(2012). 8. Kun-Ta Wu, Lang Feng, Ruojie Sha, Remi Dreyfus, Alexander Y. Grosberg, Nadrian C. Seeman and Paul M. Chaikin, “Polygamous Particles”, PNAS, 109, 18731-18736, (2012). 9. Y. Jiao, F. H. Stillinger and S. Torquato, Phys. Rev. E: Stat., Nonlinear, Soft Matter Phys., 2009, 79, 041309.? R. D. Batten, F. H. Stillinger and S. Torquato, Phys. Rev. E: Stat., Nonlinear, Soft Matter Phys., 2010, 81, 061105. “Superballs.” 10. Y.Wang, Y.Wang, D.R.Breed, V.N,Manoharan, L.Feng, A.D.Hollingsworth, M.Weck, D.J.Pine, Nature 2012, 491,51-55. 11. Lang Feng, Rémi Dreyfus, Ruojie Sha, Nadrian.C.Seeman and Paul.M.Chaikin, “DNA patchy particles”, Advanced Materials http://dx/ (2013). 12. W. B. Russel, D. A. Saville, and W. R. Showalter, Colloidal ?Dispersions, Cambridge University Press, Cambridge, England, 1989. EDITOR'S NOTE (4/07/2014): PROJECT CONTINUES AS "The Control and Dynamics of Hard Sphere Colloidal Dispersions--NNX13AR67G" WITH THE SAME PRINCIPAL INVESTIGATOR; PERIOD OF PERFORMANCE IS 9/6/2013-9/5/2018. SEE THAT PROJECT FOR SUBSEQUENT REPORTING.
Bibliography Type:	Description: (Last Updated: 09/29/2025)
Articles in Peer-reviewed Journals	Irvine WT, Bowick MJ, Chaikin PM. "Fractionalization of interstitials in curved colloidal crystals." Nature Materials. 2012 Nov;11(11):948-51. PMID: 23023553 ; http://dx.doi.org/10.1038/nmat3429 , Nov-2012
Articles in Peer-reviewed Journals	Wu KT, Feng L, Sha R, Dreyfus R, Grosberg AY, Seeman NC, Chaikin PM. "Polygamous particles." Proc Natl Acad Sci U S A. 2012 Nov 13;109(46):18731-6. http://dx.doi.org/10.1073/pnas.1207356109 ; PubMed PMID: 23100534 Polygamous particles. , Nov-2012
Articles in Peer-reviewed Journals	Feng L, Sha R, Seeman NC, Chaikin PM. "Topological interaction by entangled DNA loops." Phys Rev Lett. 2012 Nov 2;109(18):188301. PubMed PMID: 23215338 ; http://dx.doi.org/10.1103/PhysRevLett.109.188301 , Nov-2012
Articles in Peer-reviewed Journals	Palacci J, Sacanna S, Steinberg AP, Pine DJ, Chaikin PM. "Living crystals of light-activated colloidal surfers." Science. 2013 Feb 22;339(6122):936-40. http://dx.doi.org/10.1126/science.1230020 , Feb-2013
Articles in Peer-reviewed Journals	Feng L, Romulus J, Li M, Sha R, Royer J, Wu KT, Xu Q, Seeman NC, Weck M, Chaikin P. "Cinnamate-based DNA photolithography." Nature Materials. 2013 Aug;12(8):747-53. PubMed PMID: 23685865 ; http://dx.doi.org/10.1038/nmat3645 , Aug-2013
Articles in Peer-reviewed Journals	Feng L, Dreyfus R, Sha R, Seeman NC, Chaikin PM. "DNA patchy particles." Advanced Materials. 2013 May 28;25(20):2779-83. http://dx.doi.org/10.1002/adma.201204864 ; PubMed PMID: 2355415 , May-2013